

Adaptivity and Optimism

An Improved Exponentiated Gradient Algorithm

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1. Problem and Setup



1.1 What Should EG Adapt To?

Standard exponentiated gradient is robust, but its bound can be driven by experts that are irrelevant in hindsight.

Classical view

$$\text{Regret} \leq \frac{\log n}{\eta} + \eta \sum_{t=1}^T \|z_t\|_{\infty}^2$$

Desired view

$$\text{Regret} \leq \frac{\log n}{\eta} + \eta \cdot \text{complexity of } i^{\star}$$

Main question: can EG exploit that the **best expert** is easy, even when other experts fluctuate?

1.2 Expert Advice Setting

Definition (Protocol)

On each round $t = 1, \dots, T$: $w_t \in \Delta_n$ is played, $z_t \in [-1, 1]^n$ is revealed, and the learner pays $w_t^\top z_t$.

Comparator regret:

$$\text{Regret}(u) := \sum_{t=1}^T w_t^\top z_t - \sum_{t=1}^T u^\top z_t, \quad u \in \Delta_n.$$

For the best expert, take $u = e_{i^*}$.

1.3 Three Notions of “Easy”

For expert i , the paper compares:

Second moment

$$S_i := \sum_t z_{t,i}^2$$

Variance

$$V_i := \sum_t (z_{t,i} - \bar{z}_i)^2$$

Path length

$$D_i := \sum_t (z_{t,i} - z_{t-1,i})^2$$

small losses

almost constant losses

predictable from the last round

Target: depend on D_{i^*} , not D_∞ or S_∞ .

2. Motivation: Two Useful Tricks



2.1 Two Multiplicative Updates

Two updates look similar but have different guarantees:

EG / MW1

$$w_{t+1,i} \propto w_{t,i} \exp(-\eta z_{t,i})$$

Bound sees a learner-weighted or worst-coordinate scale.

MW2

$$w_{t+1,i} \propto w_{t,i} (1 - \eta z_{t,i})$$

Bound can see the best expert's second moment.

Puzzle: MW2 has the better type of bound, but it is not mirror descent with a fixed regularizer.

2.2 Trick 1: Adaptive Regularization

Write MW2 in log-weights:

$$\beta_{t+1,i} = \beta_{t,i} + \log(1 - \eta z_{t,i}).$$

Since

$$\log(1 - x) \approx -x - x^2,$$

MW2 behaves like EG plus a **second-order correction**:

$$\beta_{t+1,i} = \beta_{t,i} - \eta z_{t,i} - \eta^2 z_{t,i}^2.$$

Coordinates with larger correction terms receive stronger regularization.

2.3 Trick 2: Optimism

Suppose before seeing z_t , we have a hint m_t .

Optimistic mirror descent predicts from the preemptive update:

$$w_t = \nabla\psi^*(\theta_t - \eta m_t).$$

Regret depends on the hint error $z_t - m_t$.

If $m_t = z_{t-1}$, the error is exactly the path movement $z_t - z_{t-1}$.

3. Main Result



3.1 Adaptive + Optimistic EG

Combine the two tricks:

AEG with hints

- 1: Initialize $\beta_{1,i} = 0$.
 - 2: Predict $w_{t,i} \propto \exp(\beta_{t,i} - \eta m_{t,i})$.
 - 3: Observe z_t and suffer $w_t^\top z_t$.
 - 4: Update $\beta_{t+1,i} = \beta_{t,i} - \eta z_{t,i} - \eta^2 (z_{t,i} - m_{t,i})^2$.
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The correction penalizes experts whose hints were inaccurate.

3.2 Generic Bound With Hints

Theorem (Adaptive Exponentiated Gradient)

If $\|z_t\|_\infty \leq 1$, $\|m_t\|_\infty \leq 1$, and $0 < \eta \leq \frac{1}{4}$, then for every $u \in \Delta_n$,

$$\text{Regret}(u) \leq \frac{\log n}{\eta} + \eta \sum_{i=1}^n u_i \sum_{t=1}^T (z_{t,i} - m_{t,i})^2.$$

The comparator u chooses which coordinates matter in the second term.

3.3 AEG-Path

Choose the last loss vector as the hint:

$$m_t = z_{t-1}.$$

For $u = e_{i^*}$:

$$\text{Regret}(i^*) \leq \frac{\log n}{\eta} + \eta D_{i^*}, \quad D_i := \sum_{t=1}^T (z_{t,i} - z_{t-1,i})^2.$$

This is the path-length bound Kale asked for: dependence on the **best expert's** path length.

3.4 What Improves?

Algorithm

EG / Hedge

Cesa-Bianchi et al.

Hazan–Kale

Chiang et al.

AEG-Path

Scale in bound

S_∞

$S_{i^\star}$

$\max_i V_i$

D_∞

$D_{i^\star}$

What it misses

best expert may be easy

no variance/path adaptivity

depends on all experts

depends on worst movement

best expert only

The difference can be order T : even when $D_{i^\star} = \Theta(1)$, $\max_i D_i$ and $V_{i^\star}$ can be $\Theta(T)$.

4. Key Idea

4.1 Push Regret Into the Regularizer

The framework asks the next regularizer to absorb the one-step regret:

$$\psi_{t+1}^*(\theta_t - \eta z_t) \leq \psi_{t(\theta_t - \eta m_t)}^* - \eta w_t^\top (z_t - m_t).$$

If this holds, the regret telescopes to

$$\text{Regret}(u) \leq \frac{\psi_1^*(\theta_1) + \psi_{T+1}(u)}{\eta}.$$

The correction term is chosen to make this inequality true.

4.2 Why the Correction Has This Shape

With entropy regularization,

$$\psi^*(\beta) = \log \left(\sum_i \exp(\beta_i) \right).$$

Put $a_{t,i} = (z_{t,i} - m_{t,i})^2$. The key scalar inequality is

$$\exp(-x - x^2) \leq 1 - x \quad \left(|x| \leq \frac{1}{2} \right).$$

Therefore

$$\psi^*(\beta_t - \eta z_t - \eta^2 a_t) \leq \psi^*(\beta_t - \eta m_t) - \eta w_t^\top (z_t - m_t).$$

4.3 One Template, Different Hints

The same theorem gives different bounds by changing the hint:

Hint	Error term	Bound scale
$m_t = 0$	z_t	S_{i^*}
$m_t = \frac{1}{t} \sum_{s < t} z_s$	$z_t - m_t$	V_{i^*}
$m_t = z_{t-1}$	$z_t - z_{t-1}$	D_{i^*}

The algorithmic template is one theorem; the hint chooses the notion of predictability.

5. Significance

5.1 What the Paper Contributes

- Explains MW2 through adaptive regularizers.
- Combines adaptive regularization with optimistic learning.
- Resolves Kale's open problem: a path-length bound for the best expert.
- Extends the idea to matrix exponentiated gradient.

Limitation: adaptive tuning of η weakens the clean D_{i^*} statement.

5.2 Matrix-Valued Losses: One-Slide Extension

Replace distributions by density matrices:

$$W_t \in \mathcal{S}_+^n, \quad \text{tr}(W_t) = 1, \quad \text{loss} = \text{tr}(W_t Z_t).$$

The update becomes

$$B_{t+1} = B_t - \eta Z_t - \eta^2 (Z_t - M_t)^2, \quad W_t = \frac{\exp(B_t - \eta M_t)}{\text{tr}(\exp(B_t - \eta M_t))}.$$

With $M_t = Z_{t-1}$:

$$\text{Regret}(U) \leq \frac{\log n}{\eta} + \eta \sum_t \text{tr}(U(Z_t - Z_{t-1})^2).$$

New proof ingredient: Golden-Thompson plus FTRL-K for cone-ordered losses.

5.3 Reference

Jacob Steinhardt and Percy Liang. **Adaptivity and Optimism: An Improved Exponentiated Gradient Algorithm.** Proceedings of the 31st International Conference on Machine Learning, PMLR 32(1), 2014.

Main takeaway:

adaptive correction + optimistic hint = best-expert path-length regret

5.4 Backup: Optimizing the Path Bound

From

$$\text{Regret}(i^\star) \leq \frac{\log n}{\eta} + \eta D_{i^\star},$$

the best fixed step size is

$$\eta = \sqrt{\frac{\log n}{D_{i^\star}}}.$$

When this satisfies $\eta \leq \frac{1}{4}$,

$$\text{Regret}(i^\star) \leq 2\sqrt{D_{i^\star} \log n}.$$

If $D_{i^\star}$ is unknown, adaptive tuning is possible but the paper's clean bound becomes weaker.